

Code No: 07A1EC10

Set No. 1

I B.Tech Regular Examinations, May/Jun 2008
COMPUTER PROGRAMMING AND NUMERICAL METHODS
(Metallurgy & Material Technology)

Time: 3 hours

Max Marks: 80

Answer any FIVE Questions
All Questions carry equal marks

1. Write a C program to convert Decimal to Hexa Decimal number. [16]
2. Explain about library functions with examples. [16]
3. Explain need of pointers and its advantage. [16]
4. (a) List the differences between structure and union? [16]
 (b) How is an structure is different from an array? [8+8]
5. Define ADT . Explain with an example . [16]
6. (a) Find a positive root by Newton-Raph son's method $x \log_{10} x = 4.772393$.
 (b) Find a positive root of the following equation by bisection method $3x=6+\log_{10}x$. [8+8]
7. (a) From the following data, estimate the number of persons having income between
 I. 1000 - 1700
 II. 3500 - 4000

Income	Below 500	500-1000	1000-2000	2000-3000	3000-4000
No. of persons	6000	4250	3600	1500	650

- (b) Find $f(x)$ given the table [8+8]

X	0	1	4	5
f(x)	4	3	24	39
8. (a) Compute $\int_0^4 e^x dx$ by using Weddle's rule with 10 subdivisions
 (b) Compute the first 3 steps of the initial value problem $\frac{dy}{dx} = \frac{x-y}{2}$, $y(0) = 1.0$ by Taylor series method and next step by Milne's method with sep length $h=0.1$ also find $x(0.4)$ and $x'(0.4)$ using Adam-Bash forth method. [8+8]

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Answer any FIVE Questions
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1. (a) How to declare variable in C? What is the use of declaration?
 (b) Define constant. What are the different types of constants available in 'C'? [8+8]
2. What is the need for function? Explain about pre defined functions with an example. [16]
3. Explain need of pointers and its advantage. [16]
4. Write the definition for Structure and give the general format for declaring and accessing members. [16]
5. Explain the tower-of-hanoi problem. [16]
6. (a) Find a positive root of $3x - \sqrt{1 + \sin x} = 0$ by iteration method.
 (b) Solve for a positive root of $x e^x = \cos x$ by Regula Falsi method. [8+8]
7. (a) Given $\sin 45^\circ = 0.7071$, $\sin 50^\circ = 0.7660$, $\sin 55^\circ = 0.8192$, $\sin 60^\circ$. Find $\sin 52^\circ$, by using Newton's interpolation formula. Estimate the error.
 (b) Find the first and second difference of the polynomial $3x^4 + 8x^3 + 3x^2 - 27x + 9$. [8+8]
8. (a) Evaluate $\int_0^3 \frac{dx}{4+x^2}$ by trapezoidal rule and Simpsons 1/3rd rule.
 (b) Given $y'' + xy' + y = 0$, $y'(0) = 0$, $y(0) = 1$ Find the value of $y(0.1)$ using Runge- Kutta method of fourth order. [8+8]

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Set No. 3

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Time: 3 hours**Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

1. Explain in detail about Increment, Decrement Operators and Conditional Operators. [16]
2. (a) Write a program to show the working of static variable in different blocks.
 (b) Write a program to show the working of register variable in different blocks. [8+8]
3. Explain need of pointers and its advantage. [16]
4. Write the definition for Structure and give the general format for declaring and accessing members. [16]
5. Define ADT . Explain with an example . [16]
6. (a) Find a positive root of $xe^x = 2$ by the method of False position.
 (b) Solve the following by iteration method: $\sin x = \frac{x+1}{x-1}$ [8+8]
7. (a) Given $\sin 45^\circ = 0.7071$, $\sin 50^\circ = 0.7660$, $\sin 55^\circ = 0.8192$, $\sin 60^\circ$. Find $\sin 52^\circ$, by using Newton's interpolation formula. Estimate the error.
 (b) Find the first and second difference of the polynomial $3x^4 + 8x^3 + 3x^2 - 27x + 9$. [8+8]
8. (a) Evaluate $\int_4^{5.2} \log_e x dx$ by trapezoidal rule and Simpson's rule.
 (b) Apply the fourth order Runge Kutta method to find $y(0.2)$ given that $y' = x + y$, $y(0) = 1$ [8+8]

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Set No. 4

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Time: 3 hours

Max Marks: 80

Answer any FIVE Questions
All Questions carry equal marks

1. (a) How to declare variable in C? What is the use of declaration?
 (b) Define constant. What are the different types of constants available in 'C'? [8+8]
2. (a) Write an example to explain about call by value.
 (b) Write a factorial program using no return and no argument type. [10+6]
3. Write a program to find rank of a matrix using pointers. [16]
4. Write the definition for Structure and give the general format for declaring and accessing members. [16]
5. Define ADT . Explain with an example . [16]
6. (a) Find the positive root $x^3 - x = 1$ correct to four decimal places by bisection method.
 (b) Find the parabola of the form $y = ax^2 + bx + c$ passing through the points (0, 0), (1, 1) and (2, 20). [8+8]
7. (a) Find a polynomial which takes the following values

x	1	3	5	7	9	11
y_x	3	14	19	21	23	28

 And hence compute y_x at $x=2.12$
 (b) i. Use Lagrange's formula to find the value of y at $x = 6$ from the following data.

X	5	6	9	11
Y	12	13	14	16

 ii. Use Lagrange's formula to find y at $x = 1$ given [8+8]

X :	0	2	3	6
Y :	648	704	729	792
8. (a) A solid of revolution is formed by rotating about the x-axis, the area between the x-axis, the line $x=0$ and $x = 1$ and a curve through the points with the following co-ordinates

X:	0.2	0.4	0.6	0.8	1.00
Y:	2.0000	1.9896	1.9589	1.9089	1.8415

Estimate the volume of the solid formed using Simpsons rule.

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- (b) i. Solve $y' = x + y$; $y(0) = 1$ by Taylors series method. Find the values of y at $x=0.1$ and $x=0.2$.
- ii. Using Taylor series method compute $y(.2)$, omitting $o(h^6)$ $\frac{dy}{dx} = 1 - 2xy$ $y(0) = 0$. [8+4+4]

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